

SWE 4633

Module 1

Introduction to the Course and Cloud Computing



Course Information

- Theory/Concept + Practice for Cloud Software Development
- We use Amazon Web Service (AWS) as the main platform
- Not ready for AWS Certification, but will be helpful for preparation

Week	Mod	Week of	Activity
1	1	8/18	Introduction to the course and Cloud IaaS, PaaS and SaaS
2	2	8/25	Overview of web programming
3	3	9/1	Overview of Database
4	4	9/8	Cloud Software Development
5	4	9/15	Cloud Software Development
6	5	9/22	Cloud Computing Mechanism and Architecture
7		9/29	Group project: Independent work
8		10/6	Mid-term
9	6	10/13	RESTful APIs
10	7	10/20	Serverless Computing
11	8	10/27	Security in Cloud
12	9	11/3	Machine Learning/AI in Cloud Computing
13	9	11/10	Machine Learning/AI in Cloud Computing
14	10	11/17	More topics
15		11/24	Thanksgiving Break
16		12/1	Group project: Independent work
Final Week		12/8	Final Test

Course Information

Item	Percentage
Individual Assignments	30%
Exams	30%
Discussions	10%
Group project	30%
Total	100%

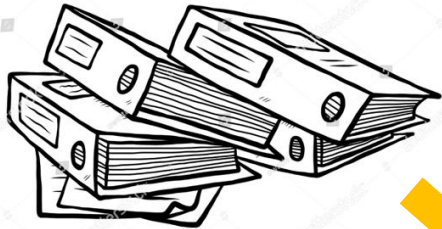


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Course Information

- Email: xli37@kennesaw.edu
- Office Location: Norton Hall, R2-204
- Office Hours: Mondy and Wednesday, 12:30pm – 3pm in the office R2-204.
 - **Please make an appointment by email before the meeting.**
- Preferred method of communication: Email

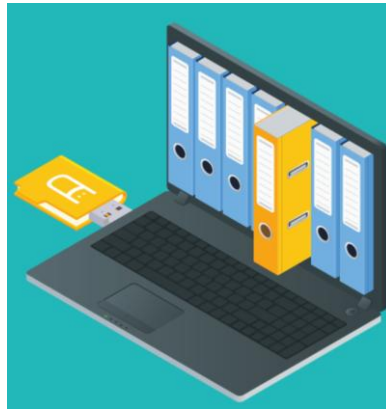
Examples of Personal Storage



Paper Document



Personal Computer



Cloud



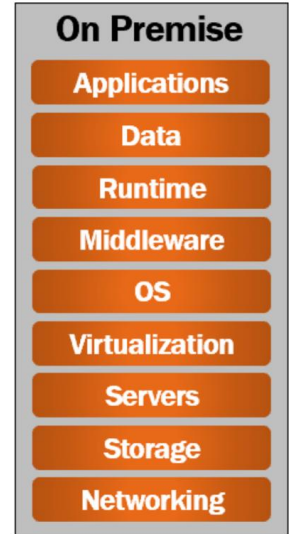
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Examples of Business Servers



On-premises IT

- On-premises software: software with the **traditional** delivery model
 - Installed and operated from an organization's **in-house** server
- Organization is responsible for managing everything:
 - Physical servers, storage and related hardware
 - Network
 - Virtualization software
 - Operating system
 - Applications
 - Middleware
 - Runtime components
- Example: make a cake at your home with everything
 - E.g., Dining table, water, electricity, ingredients, etc.



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What is Cloud Computing

- The **on-demand** delivery of IT services and resources over the Internet
 - Resources are owned and managed by the cloud provider rather than the end user
 - **On-demand:** Decision of starting and stopping service depends on customers without direct interaction with providers



Benefits of Cloud Computing

- Reduced IT costs
 - Don't need to purchase physical servers or other equipment
- Pay as you go
 - Only pay for what the organization needs and uses
- Scalability
 - Cloud provides lot of flexible and scalable options
 - Can easily scale up or scale down cloud resources like memory, processing power, storage etc.

Benefits of Cloud Computing

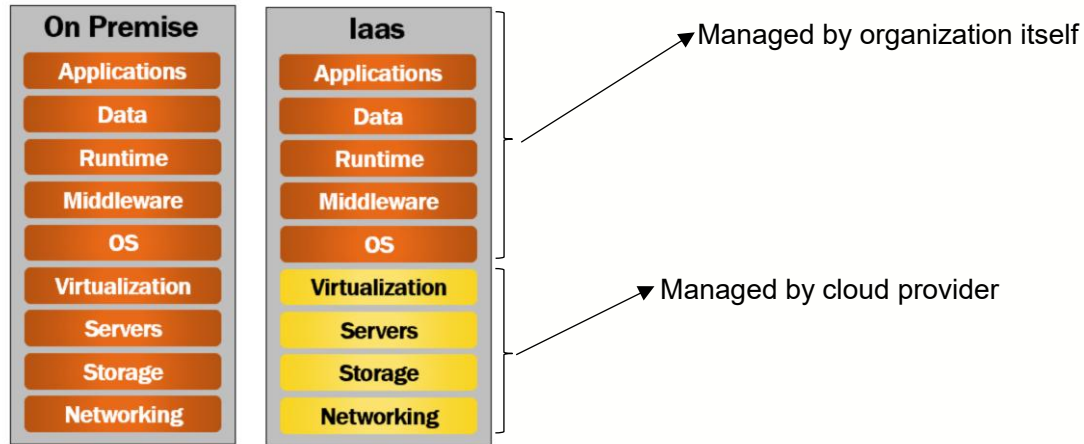
- **Accessibility**
 - Access cloud services and resources anywhere and anytime through network
- **Automatic Updates**
 - Security patches and updates are installed automatically for some of cloud services
- **Increased collaboration**
 - Team members can access edit and share files anytime and from anywhere

Risks of Cloud Computing

- Loss of cloud data and services
 - If internet connection is lost, the access to cloud data and services is also lost
- Data security
 - Many cyber attacks to the Cloud service providers
- Cost concerns
 - Cloud cost may be higher than expected
 - Make a cost plan based on the projects

Cloud Delivery Models

- Infrastructure as a Service (IaaS): like a virtual machine



- Make a cake at home but buying some ingredients
 - Such as flours, sugars, eggs, etc.

Infrastructure as a Service (IaaS)

- The most flexible cloud computing model
- Clients retain complete control of their infrastructures
- Resource purchases can be based on consumption and needs
- Highly scalable
 - IaaS allows to scale up or down to meet real-time business requirements
- Disaster recovery
 - Recovering from a natural or human-caused event

Infrastructure as a Service (IaaS)

- Security
 - Completely rely on the IaaS providers
- More training
 - Teams should get proper training to develop the required expertise, e.g., security, data backup, load balancing
- Configuration & upgrade & maintenance
 - Complete system administration and monitoring falls on user's shoulders
 - Configuration of running environment and testing platform
 - The organization is also responsible for any upgrades of software and maintenance of tools or data systems in virtual machines



Examples of IaaS



Compute Engine

Virtual Machines, Disks, Network



DigitalOcean

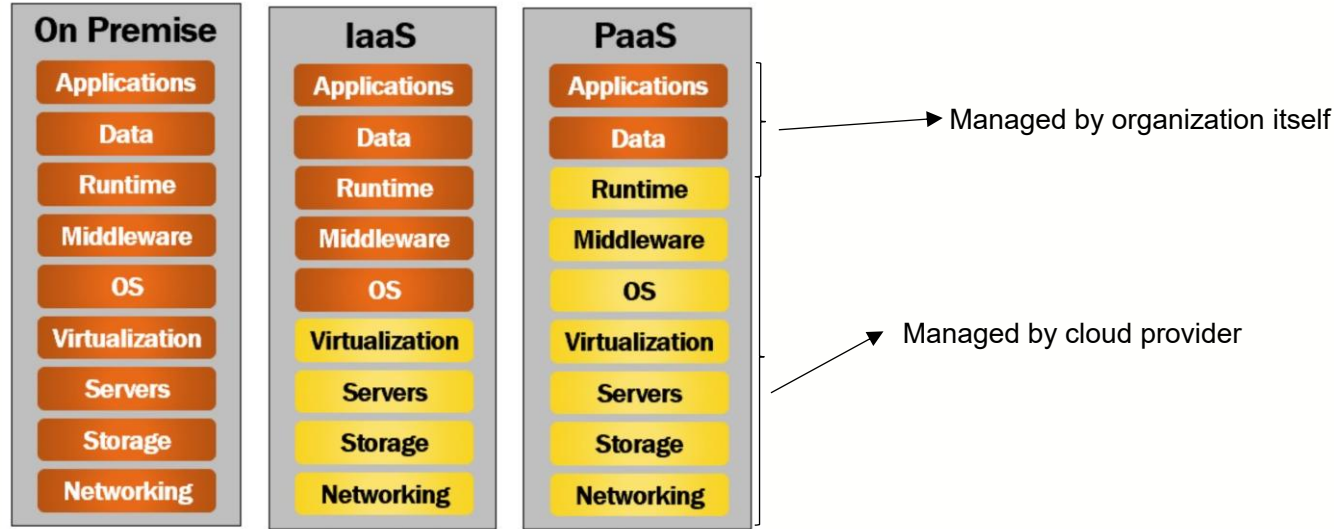


Amazon EC2



Cloud Delivery Models

- Platform as a Service (PaaS)



- Example: Order a cake, deliver to home and eat it at home

Platform as a Service (PaaS)

- Platform As a Service
 - Users can develop custom applications without having maintain all the systems needed to support the development environment
 - Focus on building the software without worrying about operating systems, software updates, storage, or infrastructure
 - Speeds up the entire development and deployment process
- Customization
 - Easily build, test, and deploy custom apps and scale your development resources up and down as needed
- The most useful model to deploy a complete application among the three models



Platform as a Service (PaaS)

- Dependency on cloud providers
 - Application is controlled by the provider's functional capabilities, speed and reliability
- Compatibility of existing infrastructure
 - A new platform is a new environment where legacy systems are supposed to continue to work
 - Migration of legacy system is not easy
- Risk of lock-in
 - Customers may get locked into a language, interface or program they no longer need

Examples of PaaS

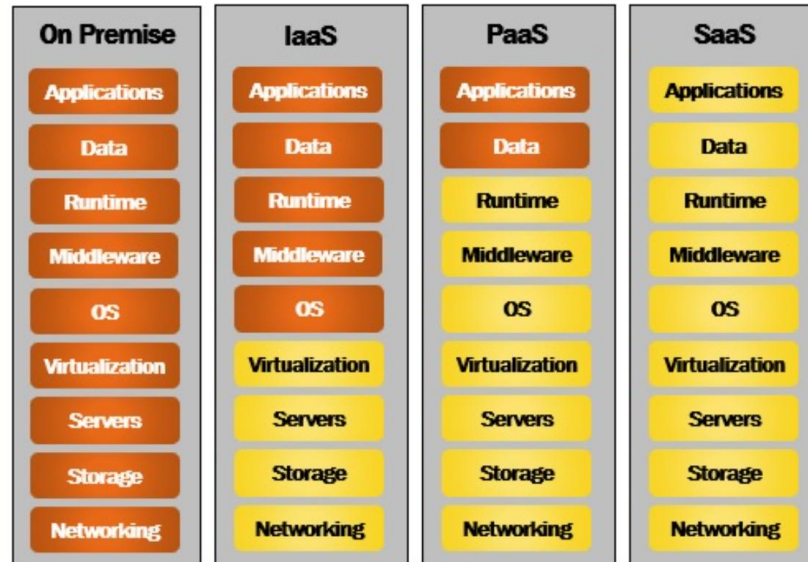


Google
App Engine



Cloud Delivery Models

- Software as a Service (SaaS)

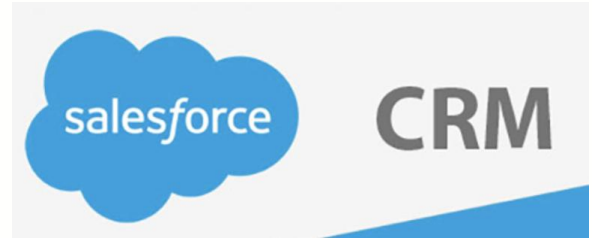


- Example: Dine out

SaaS: Software as a Service

- SaaS utilizes the internet to deliver applications, which are managed by a third-party vendor to users
- SaaS applications run directly through browsers, which means they do not require downloads or installations on the client side
- It requires vendors to manage all the technical issues
 - Customers don't need their in-house IT expertise

SaaS is Everywhere



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How to Select a Cloud Model



When we need a
virtual machine



When we need a
platform to build
a whole product



When we need to use
an application directly
from online

Other Models

- Analytics as a Service (AaaS)
 - Provides access to data analysis software and tools through the Cloud rather than having to invest in on-premise software
 - Solutions for organizing, analyzing and visualizing data
 - Even advanced techniques such as machine learning
- Database as a Service (DBaaS)
 - Use a cloud database without purchasing and setting up own hardware
- Disaster Recovery as a Service (DRaaS)
 - Disaster recovery hosted by a third party
 - Replication and hosting of physical or virtual servers to provide failover in the event of a natural disaster, power outage, or other disaster



Cloud Environments

- A **private cloud** is a **single-tenancy** environment
 - It does not share resources with other users
 - Typically resides inside the organization and must be behind a firewall
 - Only the organization has access to it and can manage it.
- A **public cloud** is a **multi-tenancy** environment
 - The data of multiple organizations is stored in a shared environment
 - Computing resources available to anyone for purchase
 - Multiple users typically share the use of a public cloud
- **Hybrid Cloud**: the cloud services with public and private clouds
 - Sensitive data kept inside the organization's network
 - Other services hosted outside the organization's network



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